Employee Burnout Risk Prediction System

INTERNSHIP PROJECT REPORT

Submitted in partial fulfillment of the requirements for the
Machine Learning Internship Program

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1. Introduction

Employee burnout has become a major concern in modern organizations. High stress levels, excessive workload, lack of sleep, and poor work-life balance can negatively affect employee productivity and well-being. Early identification of burnout risk enables organizations to take preventive actions and improve employee satisfaction.

This project aims to develop a Machine Learning model using Linear Regression to predict employee burnout risk based on workplace, productivity, and lifestyle factors.

2. Problem Statement

The Human Resources (HR) department requires a predictive system capable of identifying employees who are at risk of burnout.

The objective of this project is to build a Linear Regression model that predicts an employee's burnout risk score and identifies the factors contributing to burnout.

3. Dataset Description

The dataset contains 1000 employee records.

Features

- employee_id
- age
- years_experience
- weekly_work_hours
- meetings_per_week
- emails_sent_per_day
- projects_handled
- remote_days_per_month
- sleep_hours
- stress_level
- exercise_hours_week

- sick_leaves_year
- productivity_score

Target Variable

- burnout_risk_score

The target variable represents the burnout risk of an employee on a scale ranging from 0 to 100.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the characteristics of the dataset.

Dataset Overview

- Number of Records: 1000
- Number of Features: 13
- Target Variable: burnout_risk_score

Missing Value Analysis

The dataset was checked for missing values.

Result:

- No missing values were found in the dataset.

Statistical Summary

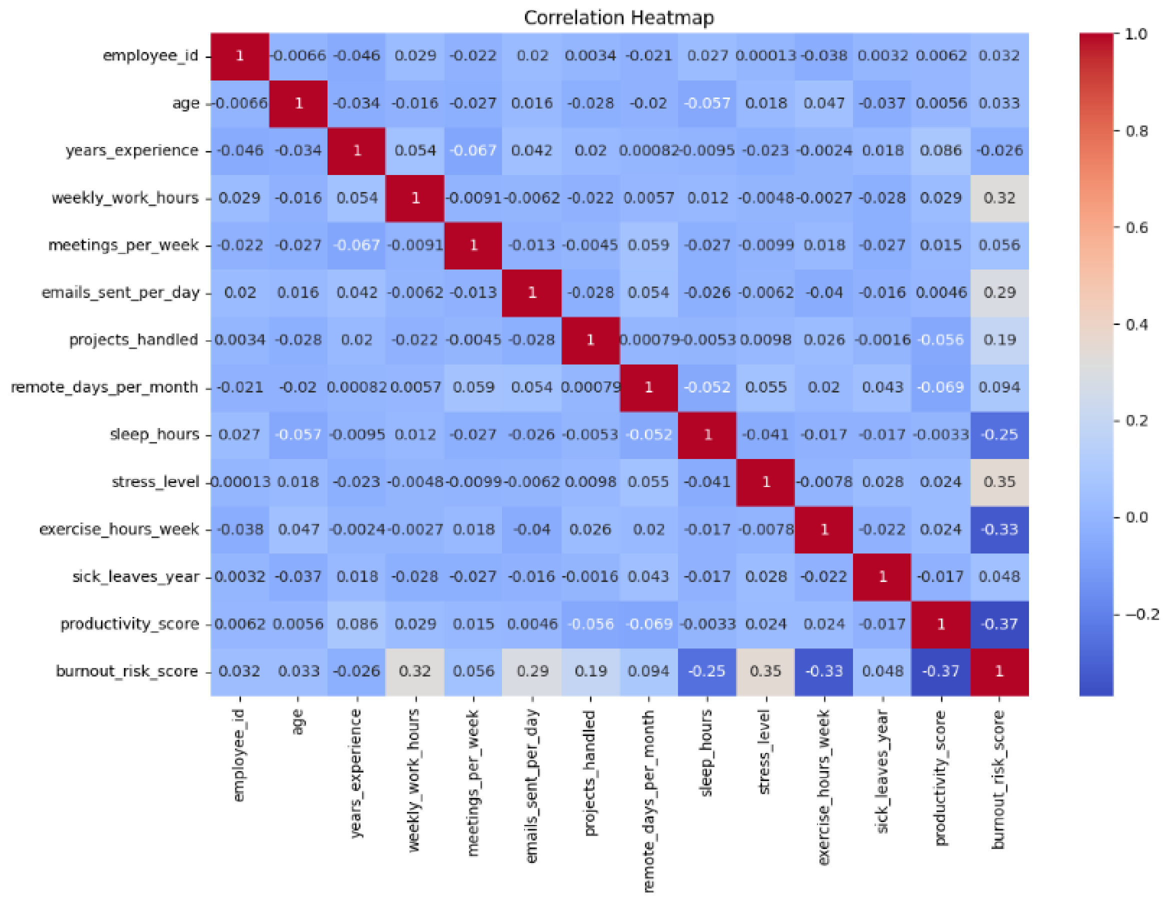
Statistical analysis was performed using descriptive statistics such as mean, minimum, maximum, and standard deviation values.

Correlation Analysis

A correlation heatmap was generated to understand relationships among variables.

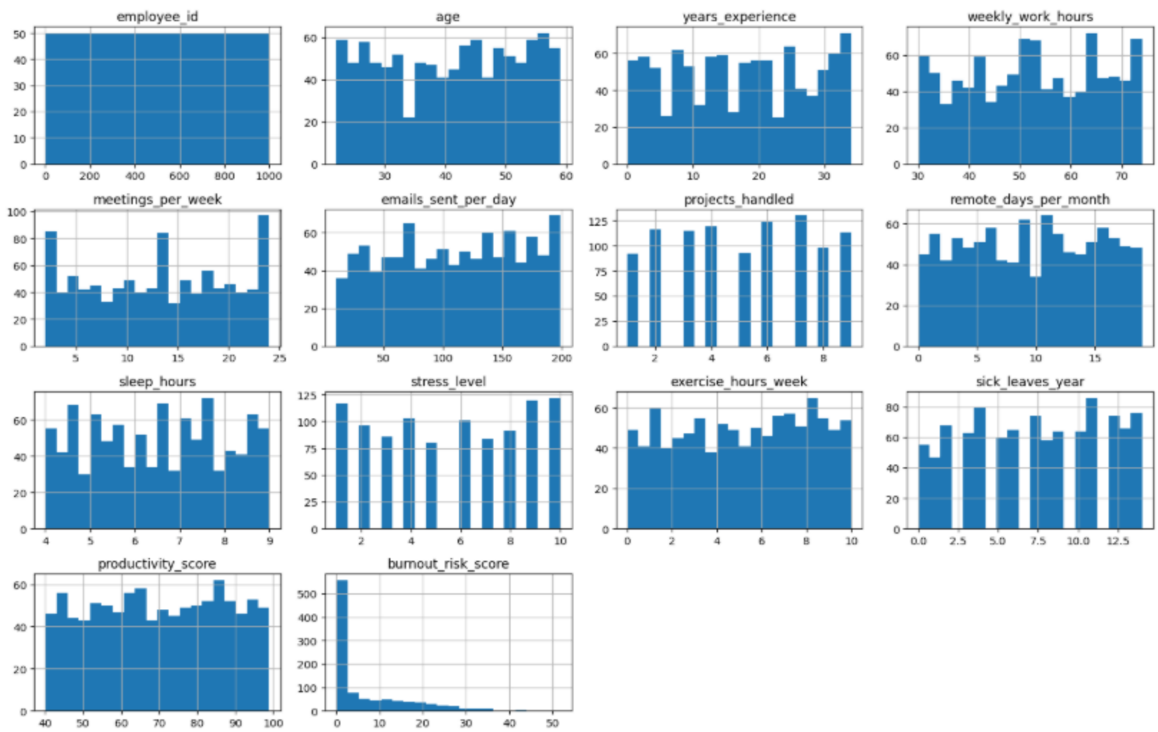
Observations

- Stress level has a positive relationship with burnout risk.
- Weekly work hours contribute positively to burnout.
- Sleep hours have a negative relationship with burnout risk.
- Exercise hours help reduce burnout risk.

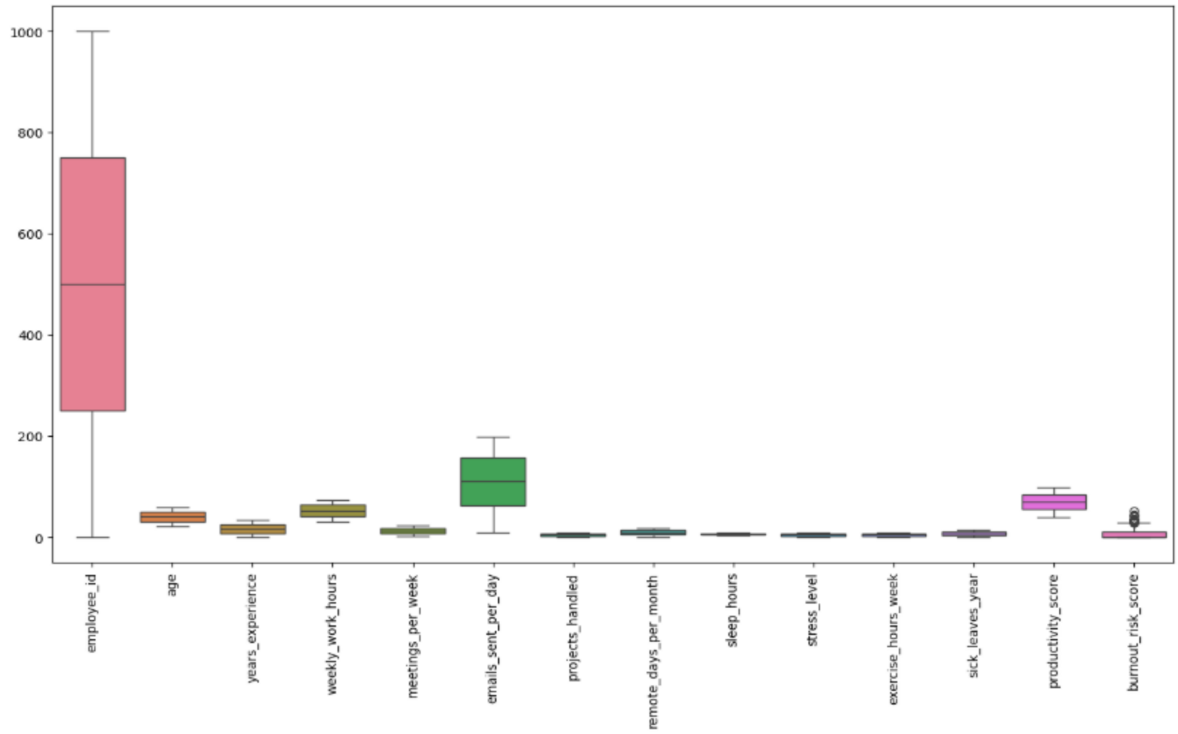


Distribution Analysis

Histograms were generated to analyze feature distributions.

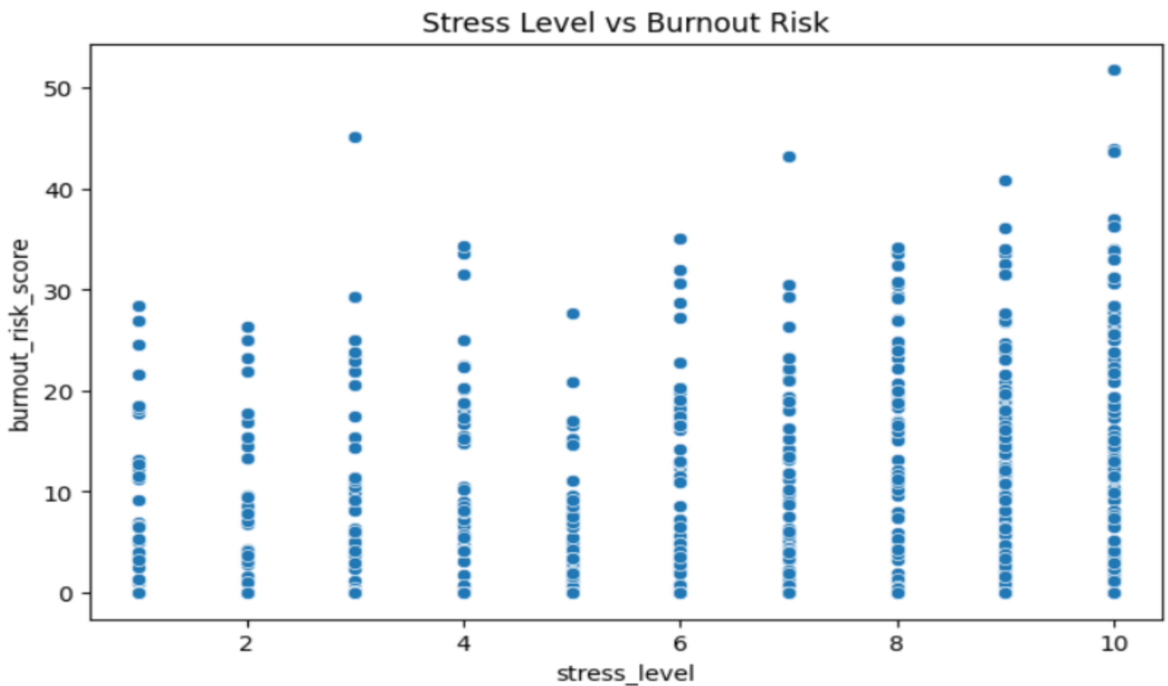


Outlier Analysis
Box plots were generated to identify potential outliers.



Scatter Plot Analysis
A scatter plot was generated between stress level and burnout risk score.
Observation:

- Higher stress levels generally correspond to higher burnout risk scores.



5. Data Preprocessing

The following preprocessing steps were performed:

1. Removed the employee_id column because it does not contribute to prediction.
2. Selected burnout_risk_score as the target variable.
3. Selected all remaining features as predictor variables.
4. Split the dataset into training and testing sets using an 80:20 ratio.
5. Prepared the data for Linear Regression modeling.

6. Model Development

A Linear Regression model was developed using the Scikit-Learn library.

Steps Performed

1. Imported Linear Regression.
2. Trained the model using training data.
3. Generated predictions on testing data.
4. Evaluated model performance.
5. Analyzed feature coefficients.

Linear Regression was selected because the target variable is continuous and the assignment specifically required this algorithm.

7. Model Evaluation

The model was evaluated using standard regression metrics.

Metric	Value
MAE	**4.65**
MSE	**34.02**
RMSE	**5.83**
R^2 Score	**0.6147**

Interpretation

Mean Absolute Error (MAE)
The model's predictions differ from actual burnout risk scores by approximately 4.65 points on average.

Mean Squared Error (MSE)
The average squared prediction error is 34.02.

Root Mean Squared Error (RMSE)
The model predictions are typically within 5.83 burnout risk points of actual values.

R^2 Score
The model explains approximately 61.47% of the variation in employee burnout risk.

8. Feature Importance Analysis

Regression coefficients were analyzed to determine the impact of each feature on burnout risk.

Major Factors Increasing Burnout

- Stress Level
- Projects Handled
- Weekly Work Hours
- Meetings Per Week
- Emails Sent Per Day

	Feature	Coefficient
8	stress_level	1.104058
5	projects_handled	0.713070
2	weekly_work_hours	0.255077
10	sick_leaves_year	0.105400
3	meetings_per_week	0.088769
4	emails_sent_per_day	0.048611
6	remote_days_per_month	0.036438
0	age	0.031162
1	years_experience	-0.004441
11	productivity_score	-0.209546
9	exercise_hours_week	-1.043501
7	sleep_hours	-1.582428

Major Factors Reducing Burnout

- Sleep Hours
- Exercise Hours Per Week
- Productivity Score

9. Business Insights

The analysis revealed several important insights:

1. Employees experiencing high stress levels are more likely to experience burnout.
2. Excessive workload significantly increases burnout risk.
3. Employees managing multiple projects are more vulnerable to burnout.
4. Adequate sleep helps reduce burnout risk.
5. Regular exercise contributes positively to employee well-being.
6. Higher productivity is associated with lower burnout risk.

10. Scenario Analysis

The trained model was used to estimate the impact of different workplace improvements.

Scenario 1: Reduce Weekly Work Hours by 10%

Result:

- Average burnout risk decreases by approximately 1.34 points.

Scenario 2: Increase Sleep by One Hour Per Day

Result:

- Average burnout risk decreases by approximately 1.58 points.

Scenario 3: Increase Exercise by Three Additional Hours Per Week

Result:

- Average burnout risk decreases by approximately 3.13 points.

Observation

Increasing exercise hours provides the largest reduction in burnout risk among the three simulated scenarios.

11. Recommendations

Based on the analysis, the following recommendations are suggested:

1. Monitor employees with high stress levels.
2. Reduce excessive workload and overtime.
3. Encourage healthy sleeping habits.
4. Promote workplace wellness and fitness programs.
5. Balance project assignments among employees.
6. Conduct periodic burnout assessments using predictive analytics.

12. Conclusion

This project successfully developed a Linear Regression model for predicting employee burnout risk. The model achieved satisfactory performance with an R^2 score of 0.6147 and an RMSE of 5.83.

The analysis identified stress level, workload, sleep habits, and exercise patterns as major factors influencing burnout. The generated business insights and scenario analysis can help HR departments proactively identify high-risk employees and implement effective burnout prevention strategies.

The proposed system demonstrates how machine learning can support data-driven decision-making in human resource management and employee well-being initiatives.